

Data Base Management System

Title Assignment - 2 (Unit - 2)

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Section - A

Course - BOTECN CSE (CORE)

Section - A

Ques 1 - Fill in the blanks:-

1. The Entity Relationship diagram visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.
2. In the Relational model, data is organised into tables, with relationships between entities define through foreign key constraints.
3. The Object Oriented model treats data as a collection of objects, each with unique attributes & methods, mirroring real-world entities and interactions.
4. Integrity constraints ensure the reliability of data by enforcing rules that must be adhere to, maintaining the accuracy of information.

Section - B

Ques 2 - How does data abstraction simplify healthcare systems?
Ans - Data abstraction hides complex database details and shows only required information to users. It allows doctors, nurses and staff to access patient records easily without understanding database structure. Different users see different views of data, which improves usability, security and efficiency in managing patient information.

Ques 3 - Explore the potential benefits and challenges associated with implementing data abstraction in this context.

Ans 3- Benefits-

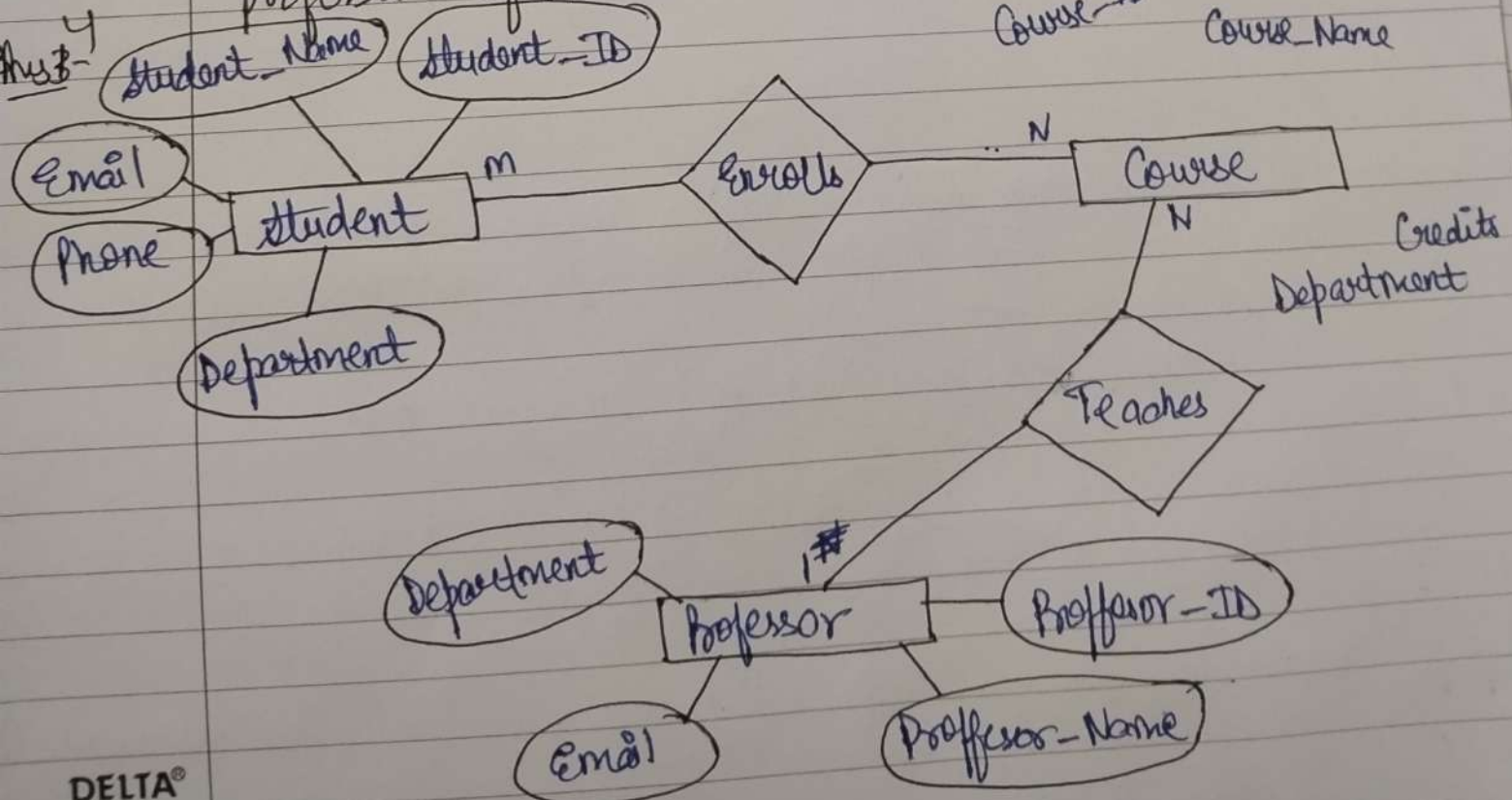
- Improves data security
- Reduces System Complexity
- Provides easy access to relevant data.
- Protects patient privacy.

Challenges-

- Difficult to design & implement
- May cause performance overhead.
- Requires proper maintenance.

Section-C

Ques 3- Design a database for a university, incorporate entities like students, Courses and professors using the entity relationship model, illustrate and explain the relationships between these entities. Include attributes for each entity to capture relevant information such as student names, course details and professor information.



→ Entities & Attributes :-

1. Student (Entity)

Attributes :-

- Student — ID (PK)
- Student — Name
- Email
- Phone
- Department

2. Course (Entity)

Attributes :-

- Course — Name
- Course — ID (PK)
- Credits
- Department

3. Professor (Entity)

Attributes :-

- Professor — ID (PK)
- Professor — Name
- Email
- Department

→ Relationships :-

1. Student — Enrolls — Course.

- A student can enroll in Courses.
- A Course can have many students.
- Many-to-Many (Relationship type).

2. Professor — Teaches — Courses
- A Professor can teach multiple courses
 - Each Course can taught by one professor.
 - One-to-many (Relationship types).

→ Explanation:-

- Student enroll in different course offered by the university
- Professors teach courses based on their specialization.
- The ER model helps visually represent the relationships between students, courses, and professors.